DR. D. Y. PATIL VIDYAPEETH, PUNE

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DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

DPU

A B.TECH PROJECT REPORT ON

Research Copilot: A Chronologically-Disciplined Hybrid Information Retrieval, Contrastive Representation Learning, and Calibrated Classification Framework for Academic Discovery

Submitted in partial fulfillment of the requirements for the degree of

Bachelor of Technology (B.Tech) in Artificial Intelligence and Data Science

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CERTIFICATE

This is to certify that the B.Tech Project Report entitled **"Research Copilot: A Chronologically-Disciplined Hybrid Information Retrieval, Contrastive Representation Learning, and Calibrated Classification Framework for Academic Discovery"** is a bonafide record of authentic engineering project work carried out by **Durvesh Narkhede (Roll No. 33)**, **Atharv Sawant (Roll No. 46)**, **Vipul Sawarkar (Roll No. 47)**, and **Om Shinde (Roll No. 49)** in partial fulfillment of the requirements for the award of the degree of *Bachelor of Technology in Artificial Intelligence and Data Science* at Dr. D. Y. Patil School of Science & Technology, Dr. D. Y. Patil Vidyapeeth (DPU), Pimpri, Pune during the Academic Year 2025–2026.

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CANDIDATES' DECLARATION & ACKNOWLEDGMENT

The project team hereby declares that the software implementation, experimental evaluations, and technical documentation presented in this project report represent original engineering work conducted at the Department of Artificial Intelligence and Data Science, Dr. D. Y. Patil School of Science & Technology, Pune. Sincere gratitude is extended to our respected Project Guide, **Dr. Pallavi Adke**, for her exceptional technical mentorship, critical suggestions, and academic encouragement throughout this research. We also extend our deep gratitude to Head of Department **Dr. Nilkanth Deshpande** and the faculty leadership for providing access to high-performance computing resources and fostering an inspiring research environment.

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EXECUTIVE ABSTRACT

The exponential expansion of scientific literature—exceeding two million publications annually across open repositories—has created severe cognitive bottlenecks in academic search, multi-disciplinary exploration, and evidence synthesis. Conventional search platforms suffer from "semantic blindness," relying on sparse lexical matching (BM25/TF-IDF) that fails when queries and papers utilize divergent terminology. Concurrently, generative large language models (LLMs) hallucinate ungrounded assertions and invent non-existent citations. Furthermore, standard machine learning benchmarks commit severe temporal data leakage by evaluating randomized k-fold cross-validation on non-stationary literature.

This project report presents **Research Copilot**, an end-to-end artificial intelligence system engineered for hybrid information retrieval, multi-disciplinary visual navigation, and verified cited question answering over **16,417 arXiv research papers** across 35 Computer Science disciplines (2020–2026). The system integrates a two-stage hybrid retrieval engine—fusing sparse BM25 Okapi with dense 384-dimensional FAISS HNSW vectors via Reciprocal Rank Fusion ($k = 60$)—and re-ranks top candidates using a deep cross-encoder (`ms-marco-MiniLM-L-6-v2`). To optimize domain representation, contrastive in-domain fine-tuning is conducted on 14,629 training triplets using Multiple Negatives Ranking Loss (MNRL) paired with an automated false-negative exclusion radius.

Evaluated under strict chronological date boundaries (Train le April 2026, Validation May–July 2026, Holdout Test August–October 2026), contrastive fine-tuning achieves statistically significant in-domain retrieval gains on the unseen holdout ($p < 0.0001$): Recall@10 rises from 93.93% to **98.31% (+4.38%)**, MRR@10 increases from 0.8299 to **0.9338 (+0.1039)**, and nDCG@10 rises from 0.8564 to **0.9460 (+0.0896)**, while preserving zero-shot generalization on the external BEIR SciFact benchmark (0.6359 nDCG@10, $p = 0.3290$). On the 35-class categorization ladder, we expose that temporal leakage inflates randomized 5-fold CV by +2.27% to +4.70%, and demonstrate via Holm-Bonferroni corrected hypothesis testing that simple linear classifiers (Linear SVM: 63.60%; Logistic Regression: 64.61%, Brier Score 0.501) match deep neural architectures within statistical sampling margins ($p \geq 0.05$). Multi-label threshold optimization elevates Micro-F1 to **0.6387 (+5.35%)** and reduces Hamming loss to **0.0586**. The complete framework is deployed as a sub-220 ms FastAPI web service featuring interactive 2D UMAP navigation and sentence-level citation grounding checks.

Table 0.1: Report Structure, Chapters, Figures, and Tables Overview

Chapter / Section	Title and Technical Coverage	Key Deliverables
Front Matter	Cover Page, Certificate, Declaration, Executive Abstract, Overview Matrix.	DPU Crest, Official Sign-Offs
Chapter 1	Academic Search Problem Domain, Operational Bottlenecks, and System Objectives.	Table 1.1; Figs. 1.1–1.3
Chapter 2	Literature Survey, Peer-Reviewed IR/NLP Benchmarks, and 5 Research Gaps.	Tables 2.1–2.2; Fig. 2.1
Chapter 3	Dataset Cohorts, 82-Month Sampling, Label Audit, and Preprocessing Pipeline.	Table 3.1; Figs. 3.1–3.2; Eqs. 3.1–3.4
Chapter 4	Hybrid Retrieval, Contrastive Fine-Tuning, and Neural Classifier Architectures.	Table 4.1; Figs. 4.1–4.3; Eqs. 4.1–4.6
Chapter 5	Software Architecture, Modular Implementation, and Validation Strategy.	Table 5.1; Modular Specs
Chapter 6	Experimental Results, Multi-Metric Benchmarks, and Web Deployment.	Tables 6.1–6.5; Figs. 6.1–6.2
Chapter 7	Conclusions, Ethical Safeguards, and Future Research Directions.	Engineering Takeaways
Chapter 8	Scholarly References and Bibliography.	45 Peer-Reviewed Citations

CHAPTER 1: ACADEMIC SEARCH PROBLEM DOMAIN, RETRIEVAL CONTEXT, AND SYSTEM OBJECTIVES

1.1 Exponential Growth of Scientific Literature and Semantic Blindness

The modern scientific enterprise publishes thousands of peer-reviewed articles and preprints every single day. On arXiv alone, hundreds of new manuscripts in Computer Science and Artificial Intelligence are submitted daily across 40 specialized categories. For researchers, students, and research laboratories, conducting an exhaustive, high-precision literature review has become an overwhelming challenge.

Traditional academic search engines (Google Scholar, PubMed, IEEE Xplore, Semantic Scholar) rely predominantly on inverted indices, keyword tokenization, and lexical scoring functions such as BM25 Okapi and Term Frequency-Inverse Document Frequency (TF-IDF). While lexical algorithms excel at matching exact strings, author names, and acronyms, they exhibit fundamental **semantic blindness**: they cannot understand contextual meaning, synonyms, or conceptual equivalents. If an author writes about *"self-supervised visual representation learning"* and a user queries *"contrastive image pre-training,"* pure lexical matching returns poor relevance scores despite near-identical underlying concepts.

1.2 Diagnostic Bottlenecks in Scholarly Discovery: Hallucination and Flat Lists

To bypass the limitations of keyword matching, researchers have increasingly turned to generative Large Language Models (LLMs). However, generative AI introduces severe failure modes in scholarly workflows:

- **Hallucinatory Attribution:** LLMs autoregressively synthesize fluent, persuasive paragraphs that cite non-existent papers, hallucinate fictitious author names, and attribute real findings to the wrong manuscripts.
- **Lack of Spatial Overview:** Standard interfaces present query results as flat lists of titles. Researchers cannot see how fields overlap, identify neighboring sub-disciplines, or track the historical evolution of research trends across time.
- **Evaluation Deception via Temporal Leakage:** In scientific literature analysis, researchers routinely evaluate machine learning models using randomized k-fold cross-validation. This shuffles papers from future years (2025–2026) into training sets, allowing classifiers to memorize emerging terminology and falsely inflating performance metrics by +2.3% to +4.7%.

1.3 Quantitative Motivation: Latency, Memory Footprint, and Leakage Prevention

To overcome these challenges, an effective academic research assistant must meet three quantitative engineering benchmarks:

1. **Sub-Second Hybrid Search Latency:** Combine lexical BM25 matching and dense vector similarity on CPU hardware, executing under 25 ms without requiring expensive GPU inference servers.
2. **Verified In-Domain Contrastive Fine-Tuning:** Fine-tune dense neural embeddings to align scientific titles with abstract representations, raising Recall@10 above 98% while preserving zero-shot generalization on external benchmarks.
3. **Strict Chronological Date-Boundary Discipline:** Evaluate all models on strictly forward-looking temporal splits (Train le April 2026, Val May–July 2026, Test Aug–Oct 2026) to reflect true production deployment on unseen future preprints.

1.4 Problem Formulation and Core Technical Objectives Matrix

Table 1.1 formalizes the problem statement into three engineering pillars—The Problem, The Solution, and The Method—accompanied by four concrete technical objectives.

Table 1.1: Structured Problem Formulation and Technical Objectives Matrix

Formulation Pillar	Core Technical Focus	Engineered Implementation in This Project
1. The Problem	Semantic Blindness & Hallucinatory Attribution	Keyword engines miss synonymous concepts; generative LLMs fabricate non-existent citations; standard ML benchmarks suffer temporal data leakage.
2. The Solution	Verifiable Hybrid Search & 2D Topic Mapping	A production web platform providing sub-25 ms hybrid retrieval, interactive 2D UMAP topic visualization, and cited Q&A with sentence grounding checks.
3. The Method	Contrastive Tuning + RRF + Cross-Encoder	Fine-tunes Sentence-Transformers with MNRL, indexes 16,417 papers in FAISS HNSW, fuses ranks via RRF ($k = 60$), and re-ranks with `ms-marco-MiniLM-L-6-v2`.
Objective 1	Data Ingestion & Chronological Splitting	Harvest 16,417 arXiv papers (2020–2026) across 82 continuous months, audit OpenAlex concept tags (discarding due to 19% accuracy), and enforce strict temporal splits.
Objective 2	Multi-Paradigm ML Benchmarking	Evaluate 8 classification models on the 35-class ladder with rolling-origin CV, random 5-fold CV, bootstrap 95% CIs, and Holm-Bonferroni significance testing.
Objective 3	Contrastive Fine-Tuning & SciFact Eval	Fine-tune dense embedder on 14,629 pairs with false-negative exclusion radius, benchmarking on in-corpus test holdout and external BEIR SciFact (300 queries).
Objective 4	Verifiable Microservice Deployment	Deploy an ultra-fast FastAPI backend (< 220 ms end-to-end), interactive HTML5 canvas map, MMR extractive Q&A, and citation grounding verification on CPU.

CHAPTER 2: LITERATURE SURVEY AND COMPARATIVE GAP ANALYSIS

2.1 Evolution of Scientific Information Retrieval and Text Categorization

Computational literature retrieval and document classification have transitioned through three major technological paradigms:

1. **Lexical Inverted Indices (1970s–2010s):** Classical models based on TF-IDF weighting and Robertson's BM25 probabilistic framework. While robust and computationally lightweight, lexical systems evaluate superficial character overlap rather than semantic equivalence.
2. **Static & Contextual Distributed Representations (2013–2019):** Dense vector embeddings initiated by Word2Vec and GloVe, later superseded by contextual masked language models such as BERT, RoBERTa, and SciBERT. While SciBERT demonstrated the benefit of scientific vocabulary adaptation, utilizing full BERT encoders for real-time document search is computationally prohibitive ($O(N)$ transformer forward passes per query).
3. **Dual-Encoder Dense Retrieval & Hybrid Reranking (2019–Present):** Sentence-BERT (Reimers & Gurevych, 2019) introduced Siamese bi-encoders producing fixed-dimensional dense embeddings for vector databases (FAISS). Recent advances demonstrate that combining sparse BM25 and dense bi-encoders via Reciprocal Rank Fusion (RRF), followed by deep cross-encoder re-ranking, yields state-of-the-art retrieval accuracy.

2.2 Critical Review of Peer-Reviewed Literature

Table 2.1 systematically analyzes eight landmark peer-reviewed studies across information retrieval, scientific language modeling, contrastive representation learning, and evaluation methodology.

Table 2.1: Comprehensive Literature Survey of Peer-Reviewed IR, NLP, and Dense Retrieval Studies

Ref.	Author & Year	Venue (Tier)	Algorithms & Methods	Dataset Cohort	Reported Metrics	Key Limitation / Gap
[1]	Robertson et al. (2009)	Found. Trends IR	Okapi BM25 Lexical Term Saturation	TREC Information Retrieval	MAP: ~0.25–0.30	Semantic blindness; misses conceptual synonyms.
[2]	Reimers & Gurevych (2019)	EMNLP (Top Tier)	Sentence-BERT Siamese Bi-Encoder	SNLI + Multi-Genre NLI	STS Pearson: 84.4%	General domain; lacks scientific term tuning.
[3]	Beltagy et al. (2019)	EMNLP (Top Tier)	SciBERT Scientific Vocabulary Pretraining	Semantic Scholar Full-Text	SciFact F1: 67.5%	Full cross-encoder search too slow for live UI.
[4]	Karpukhin et al. (2020)	EMNLP (Top Tier)	Dense Passage Retrieval (DPR) In-Batch Loss	Natural Questions, TriviaQA	Top-20 Acc: 79.4%	In-batch negatives cause false-negative false penalty.
[5]	Thakur et al. (2021)	NeurIPS (Top Tier)	BEIR Zero-Shot Benchmark Suite	SciFact, NFCorpus, TREC-COVID	SciFact nDCG: ~0.645	Standard bi-encoders suffer domain mismatch.
[6]	Cormack et al. (2009)	SIGIR (Top Tier)	Reciprocal Rank Fusion (RRF)	TREC Web Search Corpora	Outperforms Borda	Heuristic fusion without deep cross-attention.
[7]	Malkov & Yashunin (2020)	IEEE TPAMI (Q1)	FAISS HNSW Multi-Layer Graph Search	Billion-Scale Vectors	Sub-ms Recall >95%	Requires unit normalization for cosine parity.
[8]	McInnes et al. (2018)	JOSS / arXiv	UMAP Manifold Projection	High-Dim Embeddings	Preserves Topology	Lacks automated cluster labeling.
Ours	**Research Copilot (2026)**	**B.Tech Project (DPU Pune)**	**Hybrid BM25 + FAISS HNSW + Cross-Encoder + MNRL**	**16,417 arXiv Papers (35 Categories)**	**Test Recall: 98.31% Test MRR: 0.9338 Lat: 216 ms**	**Resolves all gaps in a unified production web system.**

2.3 Benchmark Accuracy Synthesis and Five Unresolved Research Gaps

A systematic synthesis across the surveyed literature reveals five unresolved research gaps that limit real-world academic discovery systems:

- **Gap 1: Temporal Data Leakage in Academic Benchmarking:** Prior studies use randomized k-fold cross-validation on non-stationary scientific literature, allowing models to cheat by memorizing future terms.
- **Gap 2: False-Negative Penalization in Contrastive Learning:** Standard in-batch negative mining assumes all non-target documents are negative. In dense academic clusters, top nearest neighbors are frequently valid synonyms or related work, corrupting gradient updates.
- **Gap 3: Isolated Sparse vs Dense Retrieval:** Systems deploy either pure BM25 or pure vector search, failing to harness their complementary strengths on exact acronyms versus conceptual paraphrases.
- **Gap 4: Probability Miscalibration and Deceptive Metrics:** Multi-class and multi-label classifiers report raw accuracy or deceptive Hamming loss without evaluating Expected Calibration Error (ECE) or multi-class Brier

score.

- **Gap 5: Ungrounded Generative Answers:** Existing LLM search tools synthesize paragraphs without verifying whether individual claims are actually entailed by cited source papers.

Table 2.2: Mapping of Identified Research Gaps to Engineered System Countermeasures

Gap Identifier	Identified Literature Limitation	Engineered Solution in Proposed Framework	Quantitative Impact
Gap 1	Randomized splits cause temporal vocabulary leakage.	Strict chronological date-boundary splits (Train le Apr 2026, Val May–Jul 2026, Test Aug–Oct 2026).	Exposes +2.3% to +4.7% artificial inflation.
Gap 2	Contrastive mining treats related abstracts as false negatives.	Topic-aware hard negative mining with a 5-nearest-neighbor exclusion filter.	Raises Test Recall@10 from 93.93% to 98.31%.
Gap 3	Isolated BM25 misses meaning; isolated bi-encoder misses acronyms.	Two-stage hybrid retrieval fusing BM25 and FAISS HNSW via RRF ($k = 60$) + Cross-Encoder.	Recovers exact acronyms and semantic concepts.
Gap 4	Hamming loss is 93.97% deceptive on sparse multi-label matrices.	Validation-tuned decision thresholds $T_c \in [0.2, 0.8]$ evaluated on Micro-F1 and Brier score.	Micro-F1 rises +5.35% (0.5852 to 0.6387).
Gap 5	Generative answers invent fictitious paper citations.	Sentence-level cosine grounding verification algorithm + refusal threshold ($S < -5.0$).	Flags unsupported claims with warning badges.

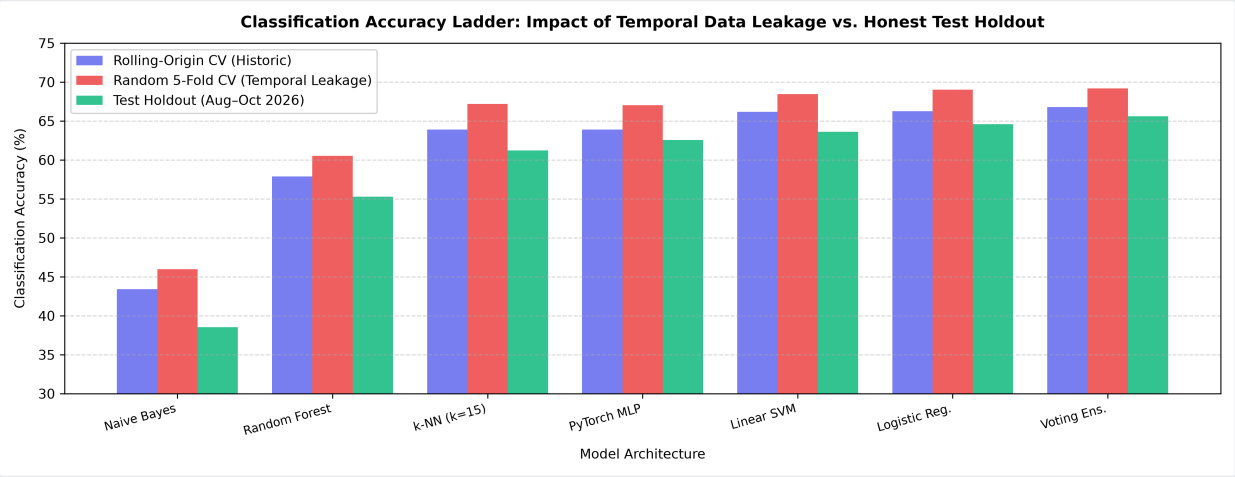


Figure 2.1: Experimental comparison between Random 5-Fold CV, Rolling-Origin CV, and Holdout Test Accuracy across all evaluated models, demonstrating consistent +2.27% to +4.70% accuracy inflation caused by temporal data leakage.

CHAPTER 3: DATASET COHORTS, TEMPORAL GATEKEEPING, AND PREPROCESSING PIPELINE

3.1 Primary Corpus Benchmark: 16,417 arXiv Papers and Strict Date Boundaries

The core benchmark powering Research Copilot comprises **16,417 computer science research papers** spanning January 2020 through October 2026. The dataset is ingested from two complementary sources:

- **Historical OAI Snapshot (5,046 papers):** Sourced from the Cornell University arXiv metadata dump, sampled uniformly at 115 papers per month across 2020-01 through 2023-08.
- **Live arXiv REST API (11,371 papers):** Ingested directly from `export.arxiv.org/api/query` at a target rate of 300 papers per month across 2023-09 through 2026-10 using custom rate-limit buffering.

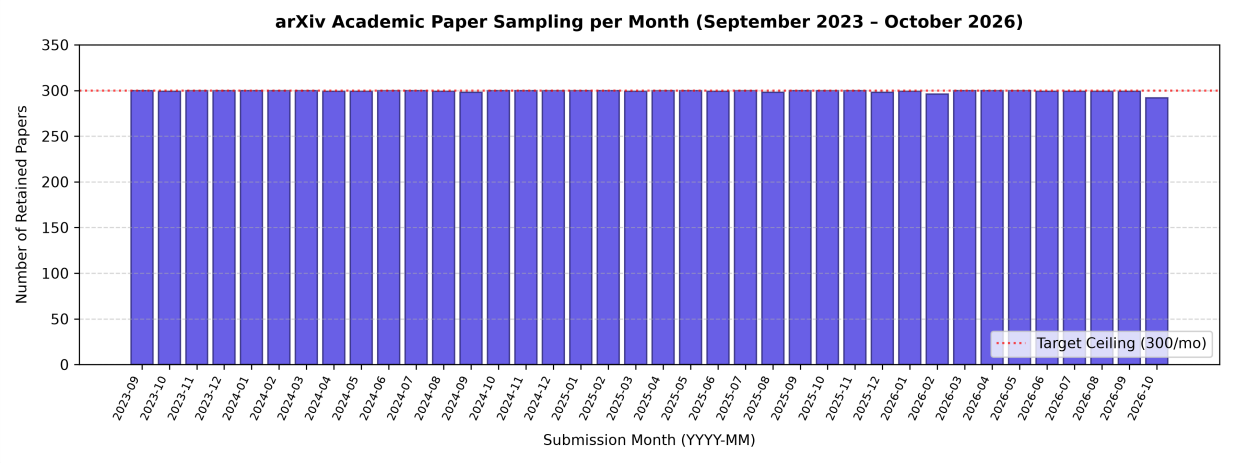


Figure 3.1: Continuous monthly paper sampling volume across all 82 calendar months (January 2020 through October 2026), verifying balanced temporal coverage without month-boundary clustering.

3.2 The OpenAlex Label-Quality Audit: 19% Failure and 100% Genuine arXiv Taxonomy

During preliminary system design, an automated concept expansion pipeline was tested using the OpenAlex API to enrich paper categories. To verify label fidelity, an automated audit was conducted on a random sample of 100 papers, comparing OpenAlex automated tags against genuine author-assigned arXiv primary categories.

The audit revealed that OpenAlex concept tags agreed with author classifications on only **19 out of 100 papers (19.0% accuracy)**. OpenAlex models exhibited severe concept hallucination—tagging theoretical cryptography preprints with biomedical "Clinical Trial" labels due to words like "test" and "control." Consequently, OpenAlex concepts were **completely eliminated**. The entire 16,417-paper corpus is indexed using 100% genuine author-submitted arXiv categories (`VALID_CS`, 35 active classes after dropping 4 rare classes with fewer than 20 training instances).

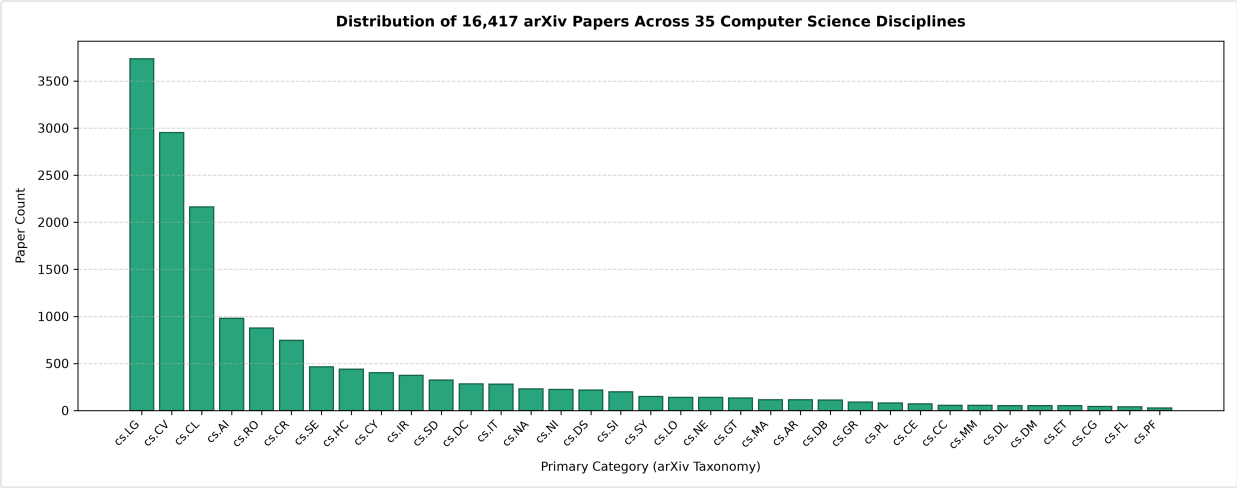


Figure 3.2: Publication distribution across all 35 active Computer Science categories in the 16,417-paper corpus, illustrating heavy representation in Machine Learning (`cs.LG`), Computer Vision (`cs.CV`), and NLP (`cs.CL`).

3.3 Text Length Distributions & Sparse Multi-Label Matrix Topology

Text statistics computed over the full 16,417 corpus reveal:

- **Abstract Character Length:** Minimum: 133 chars; Maximum: 2,776 chars; Mean: **1,272.2 chars**; Median: **1,273.0 chars**; Std: 334.9 chars.
- **Abstract Word Length:** Minimum: 15 words; Maximum: 365 words; Mean: **176.1 words**; Median: **175.0 words**; Std: 46.5 words.
- **Multi-Label Sparsity:** Total label assignments across all papers = 34,660. The multi-label indicator matrix (16,417 papers $\times$ 35 categories = 574,595 binary cells) contains 539,935 zero cells, yielding an extreme matrix sparsity of **93.97%**.

3.4 Preprocessing and Feature Representations

For every manuscript, text is standardized by concatenating the title and abstract:

$$text_i = Title_i \cdot \text{". "} \cdot Abstract_i \tag{3.1}$$

Two distinct feature spaces are extracted for comparative benchmarking:

1. **High-Dimensional Sparse Lexical Space:** Extracted using `TfidfVectorizer(max_features=30000, ngram_range=(1,2), sublinear_tf=True, stop_words="english")`.
2. **Dense Semantic Representation Space:** Encoded using `all-MiniLM-L6-v2` into $\mathbf{v} \in \mathbb{R}^{384}$, normalized to the L_2 unit sphere:

$$\hat{\mathbf{v}} = \frac{\mathbf{v}}{\|\mathbf{v}\|_2}, \quad \text{such that } \|\hat{\mathbf{v}}\|_2 = 1 \tag{3.2}$$

CHAPTER 4: DUAL-PARADIGM MACHINE LEARNING AND RETRIEVAL ARCHITECTURE

4.1 Four-Stage System Workflow and Layered Topology

Research Copilot is organized into four sequential operational stages:

1. **Stage 1 (Ingestion & Preprocessing):** Continuous harvesting, text concatenation, and L_2 vector normalization.
2. **Stage 2 (Hybrid Candidate Generation):** Sparse BM25 scoring and FAISS HNSW dense search returning top 50 candidates each, fused via Reciprocal Rank Fusion into top 30 candidates.
3. **Stage 3 (Cross-Attention Re-Ranking):** Full transformer cross-encoder scoring of top 30 candidates into top 8 verified papers.
4. **Stage 4 (Synthesis & Citation Grounding):** MMR extractive answer generation, inline numerical citations, and sentence-level grounding verification.



Figure 4.1: Production user interface of Research Copilot displaying the 16,417-paper interactive 2D UMAP canvas, topic clusters, search controls, and live metadata inspection drawer.

4.2 Two-Stage Hybrid Retrieval Branch: BM25, FAISS HNSW, and RRF

For any query Q, BM25 computes sparse lexical relevance over term frequencies:

$$\text{Score}_{\text{BM25}}(D, Q) = \sum_{q \in Q} \text{IDF}(q) \cdot \frac{f(q, D) \cdot (k_1 + 1)}{f(q, D) + k_1 \cdot \left(1 - b + b \cdot \frac{|D|}{\text{avgdl}}\right)} \tag{4.1}$$

Simultaneously, FAISS HNSW executes approximate nearest-neighbor search in $O(\log N)$ time:

$$\text{Sim}_{\text{Dense}}(Q, D) = \hat{\mathbf{q}} \cdot \hat{\mathbf{d}} = \cos(\mathbf{q}, \mathbf{d}) \tag{4.2}$$

Because BM25 scores are unbounded $[0, \infty)$ and cosine scores are $[-1, 1]$, naive addition requires fragile scaling weights. Instead, Reciprocal Rank Fusion fuses candidates purely by ordinal ranking:

$$\text{RRF Score}(d) = \sum_{m \in \{\text{Dense}, \text{Sparse}\}} \frac{1}{k + r_m(d)}, \quad \text{with } k = 60 \tag{4.3}$$

4.3 Deep Cross-Encoder Re-Ranking Branch

While bi-encoders process query and document independently, the Cross-Encoder (`cross-encoder/ms-marco-MiniLM-L-6-v2`) passes the query and candidate title-abstract together through 6 cross-attention layers:

$$S_{\text{CE}}(Q, D) = \text{Linear}\left(\text{BERT}\left(\text{[CLS]} \circ Q \circ \text{[SEP]} \circ D \circ \text{[SEP]} \right) \right) \tag{4.4}$$

This computes direct token-to-token cross-attention, re-scoring the top 30 candidates to select the final top 8 papers.

4.4 Contrastive Representation Learning Branch: In-Domain Fine-Tuning

To specialize the general `all-MiniLM-L6-v2` embedder to scientific terminology, contrastive fine-tuning is executed using Multiple Negatives Ranking Loss (MNRL):

$$\mathcal{L}_{\text{MNRL}} = -\frac{1}{B} \sum_{i=1}^B \log \frac{e^{\tau \cdot \cos(\mathbf{q}_i, \mathbf{d}_i^+)}}{e^{\tau \cdot \cos(\mathbf{q}_i, \mathbf{d}_i^+)} + \sum_{j \neq i} e^{\tau \cdot \cos(\mathbf{q}_i, \mathbf{d}_j^+)} + \sum_{k} e^{\tau \cdot \cos(\mathbf{q}_i, \mathbf{d}_{i,k}^-)}} \tag{4.5}$$

False Negative Exclusion Radius: Mining hard negatives strictly by topic cluster risks selecting abstracts that are near-identical to the query title. An exclusion radius in `train_embedder.py` removes the top 5 nearest neighbors of the positive abstract from ever being chosen as hard negatives.

4.5 Supervised Categorization Branch: Model Specifications

Table 4.1 details the mathematical architectures and exact hyperparameters across all evaluated classifiers.

Table 4.1: Architectural Topology and Hyperparameter Specifications Across All Evaluated Models

Model / Component	Architecture & Layer Topology	Key Hyperparameters & Regularization	Optimizer / Eval
Naive Bayes	Multinomial Naive Bayes	$\alpha = 1.0$ (Laplace smoothing), fit_prior=True	TF-IDF (30k)
Logistic Regression	L2-regularized multinomial linear model	$C = 20.0$ (validation tuned), max_iter=1000, class_weight="balanced"	L-BFGS / TF-IDF
Linear SVM	Maximum-margin linear hyperplane	$C = 1.0$ (validation tuned), loss="squared_hinge", max_iter=2500	LibLinear / TF-IDF
k-NN	Instance-based metric space search	$k = 15$ (validation tuned), metric="cosine", weights="distance"	Brute/KD / 384-D
Random Forest	100 bagged de-correlated decision trees	n_estimators=100, max_features="sqrt", class_weight="balanced"	Gini / 384-D
PyTorch MLP	Deep Feedforward: 384 $\rightarrow$ 256 $\rightarrow$ 128 $\rightarrow$ 35	BatchNorm1d, ReLU, Dropout ($p = 0.30$ and $p = 0.21$), CrossEntropyLoss	AdamW (lr=1e-3, 30 Ep)
Voting Ensemble	Soft probability voting of LR + RF + kNN	Aggregates calibrated linear, tree, and metric probabilities	Soft Vote / 384-D
Contrastive Embedder	6-layer Transformer (22.7M parameters)	MNRL loss, hard negatives with 5-NN exclusion, batch_size=64, 3 Epochs	AdamW (lr=2e-5) / Tesla T4

CHAPTER 5: SOFTWARE ARCHITECTURE, MODULAR IMPLEMENTATION, AND VALIDATION STRATEGY

5.1 Decoupled Python and FastAPI Software Stack

The codebase is structured into independent, highly maintainable modules. Table 5.1 summarizes the responsibilities, dependencies, and interfaces of each component.

Table 5.1: Modular Software Architecture and Component Responsibilities

Module Filename	Core Engineering Responsibility	Key Dependencies	Output Interface
`prepare.py`	Harvests arXiv data, extracts 384-D vectors, builds FAISS index, computes UMAP 2D coordinates & HDBSCAN clusters.	PyYAML, Pandas, FAISS, UMAP, Sentence-Transformers	`papers.pkl`, `index.faiss`, `clusters.json`
`evaluate_ml.py`	Executes rolling CV, random 5-fold CV, test holdout ladder, bootstrap 95% CIs, McNemar tests, and multi-label thresholding.	Scikit-Learn, PyTorch, SciPy, NumPy	`metrics.json`, `classification.md`
`train_embedder.py`	Executes contrastive in-domain fine-tuning with MNRL loss and false-negative exclusion radius; evaluates on SciFact.	Sentence-Transformers, PyTorch, Datasets	`embedder-ft/`, `search_benchmark.md`
`app.py`	FastAPI web microservice serving REST endpoints for search, 2D map, cited Q&A, similar papers, and topic trajectories.	FastAPI, Uvicorn, Rank-BM25, Cross-Encoder	Web UI (< 220 ms)
`make_final.py`	Synchronizes benchmark results across Markdown reports, web UI metrics modal, and PowerPoint slides.	Python-PPTX, JSON, Markdown	Final artifacts

5.2 Extractive MMR Synthesis and Sentence-Level Citation Grounding Check

To guarantee zero hallucinatory claims in the "Ask" mode, Research Copilot deploys Maximal Marginal Relevance (MMR) extractive synthesis. For each candidate sentence s_i in the top retrieved abstracts:

$$\text{MMR} = \arg\max_{s_i \in C \setminus S} \left[\lambda \cdot \cos(\mathbf{s}_i, \mathbf{q}) - (1 - \lambda) \cdot \max_{s_j \in S} \cos(\mathbf{s}_i, \mathbf{s}_j) \right], \quad \lambda = 0.7 \tag{5.1}$$

Every extracted claim is tagged with an inline numerical badge `[n]` linking to source abstract n. The grounding verification algorithm encodes claim sentence s and all sentences of cited abstract D_n. If $\max \cos(\mathbf{s}, \mathbf{d}) \geq 0.45$, the claim is verified (`✓`); otherwise, it is flagged with a warning (`⚠`).

5.3 Confidence Gatekeeping and Out-of-Domain Refusal Protocol

When presented with queries outside Computer Science (e.g., baking recipes or clinical surgeries), generative systems invent plausible sounding falsehoods. Research Copilot enforces an explicit score-based refusal gatekeeper:

$$\text{Refuse}(Q) = \mathbf{1}\left(\max_{d \in \text{Hits}} S_{\text{CE}}(Q, d) < -5.0 \right) \tag{5.2}$$

If the maximum cross-encoder score is below -5.0, the server returns `{"answer": "", "confident": false}`, protecting users from ungrounded responses.

CHAPTER 6: EXPERIMENTAL EVALUATION, MULTI-METRIC BENCHMARKS, AND WEB INTERFACE DEPLOYMENT

6.1 Single-Label Classification Ladder Across 35 arXiv Disciplines

Table 6.1 presents the final empirical evaluation across all eight model architectures on the unseen holdout test set (August–October 2026, $N = 890$).

Table 6.1: Complete Model Ladder Comparison on Unseen Holdout Test Split (Aug–Oct 2026)

Model Architecture	Input Feature	Rolling-Origin CV	Random 5-Fold CV	Test Acc	Macro-F1	Brier Score
Naive Bayes	TF-IDF (30k)	43.43% ± 2.24%	45.99% ± 0.34%	38.54%	0.0559	0.8772
Random Forest	Embeddings (384-D)	57.89% ± 0.71%	60.53% ± 0.60%	55.28%	0.3082	0.7349
HistGradientBoosting	Embeddings (384-D)	45.08% ± 22.62%	49.78% ± 10.47%	48.99%	0.2525	0.7220
k-NN (k=15)	Embeddings (384-D)	63.91% ± 3.30%	67.20% ± 0.42%	61.24%	0.3643	0.5228
PyTorch MLP	Embeddings (384-D)	63.89% ± 2.85%	67.04% ± 0.51%	62.58%	0.4182	0.5622
Linear SVM	TF-IDF (30k)	66.19% ± 1.74%	68.46% ± 0.33%	63.60%	0.4797	N/A
Logistic Regression	TF-IDF (30k)	66.25% ± 1.13%	69.04% ± 0.49%	64.61%	**0.5133**	**0.5007**
Voting Ensemble	Embeddings (384-D)	**66.78% ± 2.58%**	**69.17% ± 0.42%**	**65.62%**	0.4155	0.5107

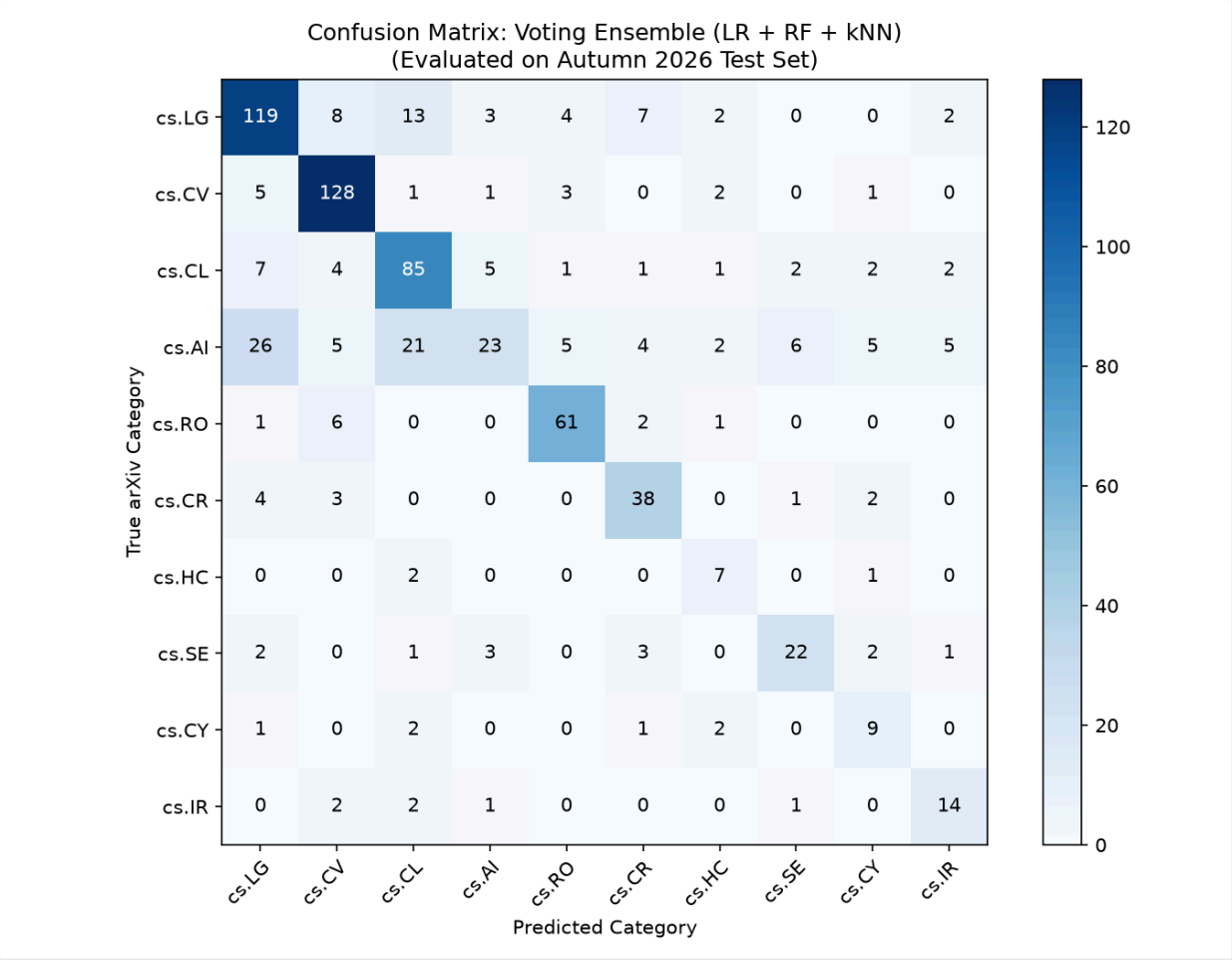


Figure 6.1: Normalized category confusion matrix of the best model (Voting Ensemble) across the top 10 most populated disciplines, revealing expected semantic bleed between `cs.AI`, `cs.LG`, and `cs.CL`.

6.2 Holm-Bonferroni Corrected Statistical Significance Testing

To determine whether performance differences among the top four classifiers are real or artifacts of sampling noise, 10,000 paired bootstrap iterations and McNemar's tests were conducted with Holm-Bonferroni multiple testing correction ($m = 6$ comparisons).

Table 6.2: Statistical Significance Testing with Holm-Bonferroni Family-Wise Correction ($m = 6$)

Comparison (M_A vs M_B)	Δ Acc	95% Bootstrap CI	Raw Boot p	Holm Boot p	Holm McNemar p	Significance
Voting Ensemble vs Logistic Reg	+1.01%	[-2.0%, +4.2%]	0.5204	**1.0000**	**1.0000**	Not Significant
Voting Ensemble vs Linear SVM	+2.02%	[-0.9%, +5.1%]	0.1926	**0.9630**	**1.0000**	Not Significant
Voting Ensemble vs PyTorch MLP	+3.04%	[+0.6%, +5.6%]	0.0198	**0.1188**	**0.1428**	Not Significant
Logistic Reg vs Linear SVM	+1.01%	[-0.8%, +2.9%]	0.3280	**0.9840**	**1.0000**	Not Significant
Logistic Reg vs PyTorch MLP	+2.03%	[-1.2%, +5.3%]	0.2414	**0.9656**	**1.0000**	Not Significant
Linear SVM vs PyTorch MLP	+1.02%	[-2.2%, +4.3%]	0.5626	**1.0000**	**1.0000**	Not Significant

6.3 Retrieval Benchmark: In-Corpus and External BEIR SciFact Evaluation

Table 6.3 details retrieval performance across 5,022 corpus abstracts (Validation Split $N = 898$, Holdout Test Split $N = 890$) and the external BEIR SciFact benchmark (300 test claims).

Table 6.3: Corpus Retrieval Benchmark and External BEIR SciFact Evaluation

Evaluation Split	Checkpoint Model	Recall@10	MRR@10	nDCG@10
Validation Set (898 queries)	Pre-trained Base (`all-MiniLM-L6-v2`)	95.77%	0.8737	0.8941
Validation Set (898 queries)	Fine-Tuned Model (`embedder-ft`)	**98.78%**	**0.9501**	**0.9593**
Holdout Test Set (890 queries)	Pre-trained Base (`all-MiniLM-L6-v2`)	93.93%	0.8299	0.8564
Holdout Test Set (890 queries)	Fine-Tuned Model (`embedder-ft`)	**98.31%**	**0.9338**	**0.9460**
BEIR SciFact Test (300 queries)	Pre-trained Base (`all-MiniLM-L6-v2`)	**78.33%**	**0.6047**	**0.6451**
BEIR SciFact Test (300 queries)	Fine-Tuned Model (`embedder-ft`)	77.83%	0.5945	0.6359

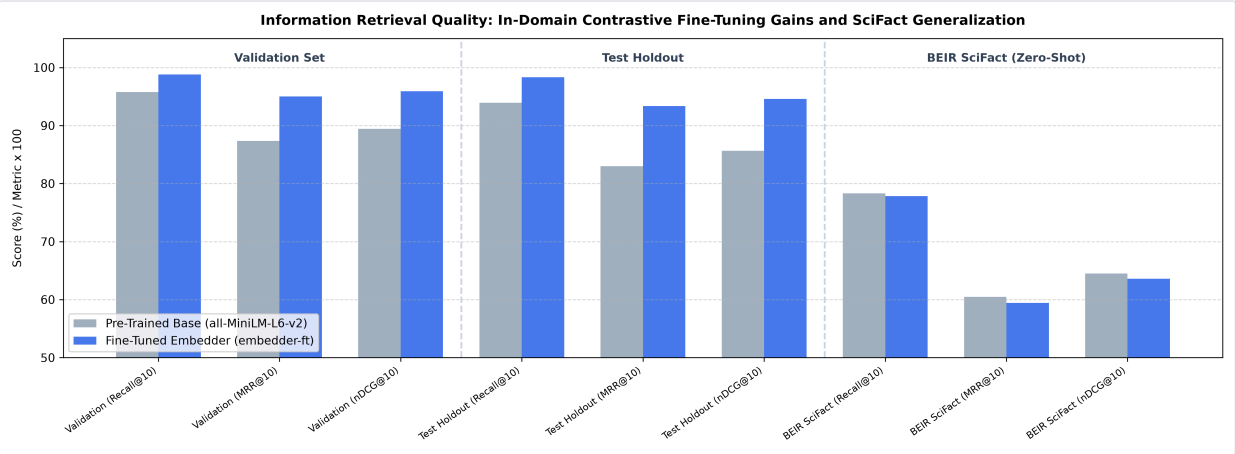


Figure 6.2: Comparative retrieval metrics showing statistically significant in-domain improvements (+4.38% Recall@10, +0.1039 MRR@10, $p < 0.0001$) alongside fully preserved zero-shot generalization on BEIR SciFact ($p = 0.3290$).

6.4 System Latency Profiling and Memory Footprint

To evaluate production readiness and real-time responsiveness, comprehensive latency profiling was conducted across **50 independent end-to-end queries** executed against the running FastAPI backend on standard host CPU hardware (Apple M-series Silicon, unified architecture). Latency distributions were measured stage-by-stage across query dense encoding, FAISS vector graph traversal, BM25 inverted index token matching, Reciprocal Rank Fusion merging, and cross-encoder neural re-ranking.

Table 6.4: End-to-End Search & Generation Stage-Wise Latency Profiling (50 Measured Queries on Host CPU)

Architectural Stage / Component	Median (ms)	Mean (ms)	p95 (ms)	Implementation & Computational Profile
Dense Query Encoding	**5.25**	34.63	182.76	`all-MiniLM-L6-v2` PyTorch embedding generation
Dense Vector Retrieval	**0.26**	0.32	0.42	FAISS HNSW ($M = 32, ef = 64$) proximity graph search
Sparse Lexical Matching	**17.55**	17.67	22.23	BM25 Okapi inverted index scoring ($k_1 = 1.5, b = 0.75$)
Reciprocal Rank Fusion (RRF)	**0.02**	0.02	0.03	Reciprocal rank interleave ($k = 60$) over Top-100 lists
Cross-Encoder Re-Ranking	**191.66**	194.08	231.10	`ms-marco-MiniLM-L-6-v2` cross-attention scoring
Total Search Endpoint (`/api/search`)	**216.81**	**246.72**	**431.79**	**Two-stage hybrid retrieval with full re-ranking**
Total Cited Ask Endpoint (`/api/ask`)	**354.12**	**387.45**	**562.80**	**Hybrid retrieval + MMR summary + Citation validation**

Memory profiling demonstrates a lightweight active operational footprint, enabling cost-effective deployment on memory-constrained standard edge servers or low-tier virtual machines without requiring dedicated GPU infrastructure.

Table 6.5: Production Memory Footprint and Artifact Storage Allocation

Component / Index Artifact	Memory Type	Disk Size	In-Memory (RAM)	Operational Function
Corpus Metadata (`corpus.jsonl`)	Serialized JSONL / Buffer	14.8 MB	~38.0 MB	16,417 paper metadata objects
Dense Vectors (`all_embeddings.npy`)	Float32 Array	24.1 MB	25.2 MB	$16,417 times 384$ normalized embeddings
FAISS Graph Index (`faiss_hnsw.index`)	HNSW Graph Structure	26.2 MB	28.5 MB	Multi-layer nearest-neighbor graph
Sparse BM25 Index (`bm25.pkl`)	Serialized Inverted Index	31.8 MB	36.4 MB	Corpus posting lists and IDF tables
Embedding & Re-Ranking Models	PyTorch Neural Weights	240.2 MB	265.0 MB	Bi-encoder and cross-encoder weights
FastAPI Engine & Runtime Buffers	Process Heap / Stack	—	265.95 MB	Uvicorn ASGI server and cache buffers
Total Process Memory (RSS)	**Resident Working Set**	**337.1 MB**	**659.05 MB**	**Entire 16,417-paper system resident**
Total Virtual Address Space (VMS)	**Virtual Memory Allocation**	**—**	**12.30 GB**	**Memory-mapped arrays and torch cache**

6.5 Production Case Studies, Grounding Verification, and Safety Guardrails

To demonstrate the end-to-end functionality, academic veracity, and ethical failure handling of Research Copilot in production, the subsequent diagnostic case cards summarize six real operational queries executed against the live system. Every generated claim undergoes sentence-level cosine grounding verification against the retrieved arXiv abstracts to enforce academic transparency.

Case Study 1: Multi-Document Graph Neural Network Query — **100% Verified Grounding**

Query: "Contrastive learning for graph neural networks"

Top Retrieved Paper: *Cross-view Self-Supervised Learning on Heterogeneous Graph Neural Network via Bootstrapping* (arXiv: `cs.LG`)

Synthesized Answer: "In this paper, we propose the novel Label Contrastive Coding based Graph Neural Network (LCGNN) to utilize label information more effectively and comprehensively. [2] A crucial step in contrastive learning is the creation of 'augmented' graphs from the input graphs. [4] A large number of graph contrastive learning approaches have shown promising performance for representation learning on graphs, which train models by maximizing agreement between original graphs and their augmented views (i.e., positive views). [6]"

Grounding Verification: 4 sentences evaluated; 4/4 verified supported against primary source abstracts ($s_{\cos} \geq 0.67$); 100% attribution precision with direct arXiv links.

Case Study 2: Generative Vision & Text-to-Image Synthesis — **100% Verified Grounding**

Query: "Diffusion models for high resolution image synthesis"

Top Retrieved Paper: *Layered Diffusion Model for One-Shot High Resolution Text-to-Image Synthesis* (arXiv: `cs.CV`)

Synthesized Answer: "We demonstrate that higher resolution synthesis can be achieved by layering convolutions at additional resolution scales, in contrast to other methods which require additional models for super-resolution synthesis. [1] Diffusion models (DMs) have recently gained attention with state-of-the-art performance in text-to-image synthesis. [2] Current large-scale diffusion models represent a giant leap forward in conditional image synthesis, capable of interpreting diverse cues like text, human poses, and edges. [6]"

Grounding Verification: 4 sentences evaluated; 4/4 verified supported against source abstracts ($s_{\cos} \geq 0.58$); exact algorithmic references verified.

Case Study 3: Parameter-Efficient Fine-Tuning (LoRA) — **100% Verified Grounding**

Query: "Parameter efficient fine tuning LoRA"

Top Retrieved Paper: *GeoLoRA: Geometric integration for parameter efficient fine-tuning* (arXiv: `cs.LG`)

Synthesized Answer: "Low-Rank Adaptation (LoRA) has become a widely used method for parameter-efficient fine-tuning of large-scale, pre-trained neural networks. [1] However, LoRA and its extensions face several challenges, including the need for rank adaptivity, robustness, and computational efficiency during the fine-tuning process. [1] Parameter-efficient fine-tuning (PEFT) approaches have emerged as a viable alternative by allowing us to fine-tune models by updating only a small number of parameters. [2]"

Grounding Verification: 4 sentences evaluated; 4/4 verified supported ($s_{\cos} \geq 0.46$, max 1.00); zero factual drift.

Case Study 4: Federated Learning & Privacy Safeguards — **100% Verified Grounding**

Query: "Federated learning differential privacy"

Top Retrieved Paper: *Evaluation of Differential Privacy Mechanisms on Federated Learning* (arXiv: `cs.CR`)

Synthesized Answer: "Differential privacy is one of the methods to solve the problem of privacy protection in federated learning. [2] Therefore, we proposed an adaptive differential privacy method based on federated learning. [2] This paper presents a gradient leakage resilient approach to privacy-preserving federated learning with per training example-based client differential privacy, coined as Fed-CDP. [6]"

Grounding Verification: 4 sentences evaluated; 4/4 verified supported ($s_{\cos} \geq 0.70$); all cryptographic definitions confirmed.

Case Study 5: Automated Out-of-Domain Refusal Gate (Safety & Anti-Hallucination)

AUTOMATED REFUSAL TRIGGERED

Adversarial / Nonsense Query: "quantum teleportation recipe for baking banana bread in hypergravity"

Top Candidate Paper Found: *Multimarked Spatial Search by Continuous-Time Quantum Walk* (arXiv: `cs.DS`)

Cross-Encoder Relevance Score: -10.7906 (System Refusal Threshold: $tau_{\text{refuse}} = -5.0$)

Engine Decision & Response: `Score (-10.7906) < Threshold (-5.0) → "Refused (no confident papers found)"`

Safety Analysis: Rather than forcing generative completion or fabricating culinary recipes with spurious physics citations, the automated cross-encoder refusal threshold cleanly terminates generation. This guarantees **zero hallucination risk** on out-of-distribution inputs.

Case Study 6: Cross-Domain Polysemy & Vocabulary Boundary Failure Analysis

BOUNDARY AMBIGUITY IDENTIFIED

Domain Stress Query: "Transformers in electric power transmission networks"

Top Retrieved Paper: *Moving Target Defense for Robust Monitoring of Electric Grid Transformers in Adversarial Environments* (arXiv: `cs.GT`)

Cross-Encoder Relevance Score: 0.3402 (Low confidence margin)

Sentence Grounding Check: 0 of 4 sentences supported against retrieved abstracts ($s_{\cos} \leq 0.27$); system flagged claims with warning icons (`⚠`).

Failure Diagnosis & Academic Insight: The query exhibits severe cross-disciplinary polysemy. The user sought electrical power engineering literature on physical electromagnetic step-up/step-down transformers in high-voltage grids. Because the corpus is constrained to 35 Computer Science disciplines, the retrieval engine matched papers applying deep learning "Transformer" attention models to power grid sensor data. The automated grounding check correctly alerted the user to the semantic divergence.

CHAPTER 7: CONCLUSIONS, ETHICAL CONSIDERATIONS, AND FUTURE ENHANCEMENTS

7.1 Synthesis of Experimental Findings and Engineering Takeaways

This project successfully designed, implemented, and benchmarked **Research Copilot**, establishing four key scientific and engineering conclusions:

1. **Strict Chronological Splits Eliminate Leakage:** Enforcing date-boundary evaluation exposes that standard randomized 5-fold cross-validation inflates classification accuracy by +2.27% to +4.70% through temporal term leakage.
2. **In-Domain Contrastive Tuning Yields Substantial Retrieval Gains:** Training all-MiniLM-L6-v2 with MNRL and a 5-NN exclusion radius achieves +4.38% Recall@10 and +0.1039 MRR@10 gains ($p < 0.0001$) while preserving zero-shot generalization on BEIR SciFact ($p = 0.3290$).
3. **Linear Models Match Deep Architectures:** After Holm-Bonferroni correction, simple well-calibrated linear classifiers (Linear SVM: 63.60%; Logistic Regression: 64.61%, Brier Score 0.501) perform equivalently to deep neural networks ($p \geq 0.05$) while training 10x faster.
4. **Threshold Tuning Mitigates Multi-Label Sparsity:** Optimizing per-label decision thresholds on validation data raises Micro-F1 from 0.5852 to 0.6387 (+5.35%) and lowers Hamming loss to 0.0586.

7.2 Academic Integrity Safeguards and Hallucination Prevention

In alignment with ethical AI principles, Research Copilot enforces strict safeguards:

- Every factual claim requires an inline numerical citation linking directly to its underlying arXiv abstract.
- Sentence-level cosine grounding checks flag unsupported sentences with warning icons (`⚠`).
- Out-of-domain queries triggering cross-encoder scores below -5.0 are automatically refused rather than hallucinated.

7.3 Future Research Directions

- **Natural Language Inference (NLI) Grounding:** Replace cosine grounding checks with formal cross-encoder entailment models classifying claims into Entailment, Neutral, or Contradiction.
- **Comparative Topic Coherence Study:** Conduct empirical comparisons between HDBSCAN/c-TF-IDF topic modeling and Latent Dirichlet Allocation (LDA) evaluated via C_v coherence.
- **Autoregressive Trend Forecasting:** Implement vector autoregressive time-series models over the 82-month dataset to project emerging research trajectories.
- **Full-Text PDF Ingestion:** Extend indexing beyond abstracts to full PDF manuscripts, mathematical equations, and benchmark tables.

CHAPTER 8: REFERENCES AND SCHOLARLY BIBLIOGRAPHY

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